Literature Survey for IoT-based Smart Home Automation: A Comparative Analysis

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***Abstract*— Any technology that allows a gadget to connect to the Internet is referred to as the "Internet of Things." For such systems, data collection is crucial. Afterward, the data is used for internet-based control, monitoring, and data transmission to other devices. Smart House technology is the future of domestic technology, with the goal of offering and distributing a range of services both within and outside the home via networked devices that combine and integrate all of the many applications and intelligence behind them. These smart gadgets have the capacity to communicate with one another thanks to the continual availability of a broadband internet connection. As a result, smart home technology is now integrated into the Internet of Things (IoT). We provided a literature review on Smart Home Automation in this research study, which gives homeowners peace of mind by allowing them to monitor and protect their houses from away.**

Keywords*— *Internet of things (IoT), Smart Homes, Arduino, Blynk, Automation, Sensors.

I. INTRODUCTION

A smart house is one in which appliances can be controlled and monitored remotely via internet-connected gadgets. Smart home technology allows homeowners to operate smart appliances from their smartphones or other connected devices, increasing security, comfort, convenience, and energy savings. Devices and systems for smart homes that are connected to the internet of things (IoT) frequently interact, exchanging data on consumer usage and automating operations depending on the preferences of the homeowners. Home automation provides homeowners peace of mind by allowing them to keep a watch on their houses from afar and avoid risks such as an unlocked front door or a forgotten coffee maker. Users' convenience needs may be addressed in smart homes. Consumers can be more productive with home automation. Rather than operating the air conditioner all day, a smart home system might learn how people live and act in order to keep the house cool when they get home from work. Grass will only be watered when a smart irrigation system requires it, and only with the exact amount of water necessary, thanks to IoT-based home automation. Customers may save money and natural resources by using less electricity, water, and other resources with home automation. A smart home is more than a collection of smart devices and appliances; it's a system of networked objects that can be managed from anywhere. A master home automation controller, often known as a smart home hub, is in charge of controlling all of the gadgets. A smart home hub is a physical device that connects to the internet and acts as the brain of the system, detecting, processing, and transferring information wirelessly. It brings together all of your smart home apps into a single app that customers can use from anywhere. We introduced Smart Home automation based on IoT utilizing the Arduino & Blynk Application in the first section of this research paper. The literature review of work done by various authors in the field of Smart Home automation based on IoT utilizing the Arduino & Blynk Application is presented in the second section of the research study. We offered a comparative assessment of the work done by many authors in the field of Smart Home automation based on IoT utilizing the Arduino & Blynk application in the third and fourth sections of this research paper and presented their work in a tabular style. Finally, we found that Internet-of-Things-based Home Automation solutions have several advantages over wired systems, including ease of use, ease of installation, avoidance of the complexity of running through cables or weak electrical connections, problem detection and triggering, and, most importantly, mobility.

II. RELATED WORK

N. Ganesh [1] (2017) has given a GSM Technology Implementation of IoT Architecture for Smart home that offers a system that would allow users to operate and monitor smart gadgets through the internet. The given system provides a web-based IoT architecture based on GSM for the deployment of smart home apps, as well as a GSM-based smart house control system design. In the given system Data transmission and authentication, processes need to be made more secure.

Laila S., et al [2] (2016) have given an IoT-Based Smart Home with Low Energy Consumption where an energy-efficient IoT-based smart house is demonstrated using a home model. In the proposed system Antenna modules are unique to each wireless device with having the individual Antenna module on a smart LED, as well as one on the HVAC system.

Nasir N.K., et al [3] (2020) has given IoT Implementation Using Android SDK and Cayenne Web Server. The proposed system offers a smart home implementation that can regulate the house's atmosphere by analyzing customer behavior using data acquired from numerous sensors. In this deployment of a smart house that focuses on energy conservation. Further in it, the Cameras may be utilized to keep an eye on what's going on inside the house and provide a live stream. As future work, it is also possible to automate landscape watering systems.

Guo-Ming S., et al [4] (2019) have given a Smart home system based on IoT with a mobile phone app and a virtualized cloud server. In the proposed system a mobile phone app with a virtualized cloud server is used to power an IoT-based smart home system. The merit of it is that to finish data analysis and data management fast and properly, the suggested system blends software and hardware.

Priya N., et al [5] (2018) have proposed their work Using the Smart Door Authentication Algorithm, for secure smart home automation based on IoT by putting in place a mechanism to assist the elderly and disabled. In the proposed system the layout of the project is straightforward.

Shivani J., et al [6] (2020) have proposed their work for a Comfortable Smart Home by using an IoT-based home automation system to connect households and appliances by using technologies such as the IoT and Machine Learning in a cost-effective manner to make them smart. In their proposed

system users will be able to get timely message notifications thanks to the mobile application's connection. As future work, the features like Voice assistants, emergency triggers, and music/tone recognition controls for physically challenged persons may all be added to the system.

Satyendra K. V., et al [7] (2019) have proposed a home automation system that is both clever and energy-efficient. This article proposes a home automation system that is both clever and energy-efficient that has access to and can operate household appliances from anywhere on the planet. As a future work the suggested system needed to be tested before it could be used in a real-world setting.

Anitha A., et al [8] (2017) have proposed their work on IoT, where it is being used to create a home security system. The proposed technology may be employed in both industrial and commercial settings. For future work, an RFID scanner may be added to the system to further enhance it.

Shubham N. S., et al [9] (2018) have proposed an Arduino-based IoT Home Automation System in which a mobile host controller and numerous client modules are included in this home automation system. Using the proposed system People can automate the control of their electrical appliances and set up the remote control. As a disadvantage, this system does not provide multiple appliance control or automated fault detection in the controlled appliance.

Mohammad A., et al [10] (2019) have proposed a Smart device-based IOT home networking and architectural design connection between machines. The proposed system is Wi-Fi and Web-based system that uses the WSN nodes along with Raspberry pi. In the proposed system a web browser may control the entire system.

Amr Jadi, et al [11] (2020) has proposed their work using IoT devices, where a strategy for predicting uncertainty in the home environment is suggested. They have used Sensors (IoT devices) for forecasting uncertainty using a neural network-based system that is suggested as a novel technique for predicting unpredictable events in the home environment.

Lakshmi, et al [12] (2018) have proposed their work on implementing a Smart Home Security System with Door Alert Powered by IoT, in which they have used Raspberry Pi to control illegal entry to our doors. In the proposed work unauthorized entrance to homes and workplaces is prevented using a Raspberry Pi connected to a web camera and a PIR motion sensor. The proposed system has to be validated and tested on a wide scale.

Nazmul H., et al [13] (2018) have proposed a Smart Home Security Framework which is based on IoT which uses the Home automation system based on Cisco's IoT security architecture. In the proposed system the home automation system may connect through the internet and employs portable devices as a user interface.

Lalit M. S., et al [14] (2018) have proposed Home automation using Arduino and the IoT which uses an Arduino microcontroller to create a low-cost, adaptable, and dependable home automation system with added security. In the proposed work Any Wi-Fi network can be used to govern the proposed wireless network. The proposed system has to be validated and tested on a wide scale.

Poongothai, et al [15] (2018) have proposed their work for the implementation of an Internet-of-Things (IoT)-based Smart Laboratory. The main goal of the proposed work is to make electrical appliances more accessible to users through an Android app and a Node-RED dashboard. The advantage of the proposed work is that the users may operate equipment such as lights, fans, projectors, and air conditioners simply by providing orders via a smartphone or the Node-RED Dashboard.

Chakib B., et al [16] (2014) have proposed their work Smart Grid with IoT in concern to look at the IoT-based SG's security concerns and challenges. In their proposed work using Zigbee and Raspberry pi networks, they have examined the SG's key security concerns and problems.

Pranay P. D., et al [17] (2018) have proposed their work on IoT-based smart home automation security where they worked mainly to regulate other electrical appliances and monitor the temperature in the house. The advantage of their proposed work is that by using a web-based mobile app and the ARDUINO-UNO (microcontroller) the IoT-based Wi-Fi has been successfully tested on a variety of smartphones, demonstrating its mobility and adaptability.

Kunal D., et al [18] (2016) have proposed their work using a Home security automation based on the IoT to assist the disabled and elderly. They have proposed that Using Home security automation IoT based systems are designed to help the aged and handicapped. The merit of their work is that the suggested solution is both secure and energy-efficient.

Dhanusree, et al [19] (2020) has used Wireless security Internet-of-things-based smart home automation to propose a wireless home automation system for managing the features and functions of home appliances using the computer and mobile devices. By using a web-based mobile app and the Arduino microcontroller users can use their mobile devices to control their electrical equipment by programming actions.

Jitendra J., et al, [20] (2017) have proposed their work on Smart Home Performance Enhancement and Monitoring through the IoT. They have suggested a hierarchical smart home service architecture based on common home interface devices by using a Wi-Fi network, web-based mobile app, and NRF modules which are incorporated into Raspberry Pi and Arduino. The advantage of this proposed work is that High-tech smart home systems are intended to be converted into simple and easy-to-use home automation systems through the use of the suggested hierarchical design.

III. A COMPARATIVE ANALYSIS OF THE RELATED WORK

In this section, a comparative review of the related work is carried out and is given in Table 1 in a tabular format.

IV. A TABULAR ANALYSIS OF THE COMMUNICATION INTERFACES USED

In this section we have provided a technical comparative review of the various technical communication interfaces related work is carried out and is given in Table 2 in a tabular format.

V. CONCLUSION

We present a comparative analysis of the literature study of systems employing the Internet of Things in smart home automation applications in this research paper. We attempted to address gaps in smart home automation using IoT by overcoming the challenges that people face daily in a society where recurrent power outages, uncontrolled urbanization, a labor shortage in agriculture and farming, and other issues are all too evident in the implementation of remote system monitoring and control. We've also seen that Internet-of-Things-based Home Automation solutions have several advantages over wired systems, including ease of use, ease of installation, the elimination of the complexity of running through cables or weak electrical connections, problem detection, and triggering, and, most importantly, mobility.

REFERENCES

[1] E.N.Ganesh, et al.,"Implementation of IOT Architecture for SMART HOME using GSM Technology", International Journal of Computer Techniques – Volume 4 Issue 1, Jan – Feb 2017.

[2] Laila Salman, et al.,"Energy efficient IoT-based smart home", pub. in proc. of IEEE 3rd World Forum on Internet of Things (WF-IoT), 2016.

[3] Nasir Nazir Khan, et al.,"IoT-Based Smart Home Implementation Using Cayenne Web Server and Android SDK", International Journal of Advanced Science and Technology Vol. 29, No.4, pp. 4034 – 4041, 2020.

[4] Sung G-M, et al.,"Internet of Things–based smart home system using a virtualized cloud server and mobile phone app.", pub. In Int. Jour. of Distributed Sensor Networks., September 2019.

[5] Priya Narke, et al.,"IoT based smart security and Home Automation", pub. in proc. of Int. Conf. on recent development in Science, Engineering, Management & Humanities (SEMH 2018), Institution of Engineers, Mumbai, 2018.

[6] Shivani Jadon, et al.,"Comfy Smart Home using IoT", pub. in proc. of International Conference on Innovative Computing and Communication (ICICC-2020), 2020.

[7] Satyendra K. Vishwakarma, et al.,"Smart Energy Efficient Home Automation System Using IoT", pub. in proc. of 4th IEEE International Conference on Internet of Things: Smart Innovation and Usages (IoT-SIU), 2019.

[8] Anitha A, et al.,"Home security system using internet of things", pub. in proc. of 14th ICSET-2017, IOP Conf. Series: Materials Science and Engineering, 263, 2017.

[9] S. H. Dinde, et al.,"IoT Based Home Automation System Using Arduino", International Journal of Advanced in Management, Technology and Engineering Sciences, Volume 8, Issue IV, APRIL, 2018.

[10] Mohammad Ayyub, et al.,"IoT based Intelligent Home Using Smart Devices", Complexity International Journal (CIJ), Volume 23, Issue 2, PP:178-185, July-August 2019.

[11] Amr Jadi, et al.,"Place-based Uncertainty Prediction using IoT Devices for a Smart Home Environment", (IJACSA) International Journal of Advanced Computer Science and Applications, Vol. 11, No. 3, 2020.

[12] S. Lakshmi Ojaswini, et al.,"IoT based Smart Home Security System and Door Alert using Smart Phone", IRE Journals, Volume 1, Issue 8, Feb. 2018.

[13] Nazmul Hossain, et al.,"A Security Framework for IOT based Smart Home Automation System", Global Journal of Computer Science and Technology: E Network, Web & Security, Volume 18 Issue 3 Version 1.0 Year 2018.

[14] Lalit Mohan S., et al., "Arduino based home automation using Internet of things (IoT)", International Journal of Pure and Applied Mathematics Volume 118 No. 17, 769-778, 2018.

[15] M. Poongothai, et al.,"Implementation of IoT based Smart Laboratory", International Journal of Computer Applications, Volume 182 – No. 15, September 2018.

[16] Chakib Bekara, et al.,"Security Issues and Challenges for the IoT-based Smart Grid", Procedia Computer Science, Volume 34, Pages 532-537, 2014.

[17] Pranay Pratim Das, et al.,"Smart Security & Home Automation Using Internet of Things (IoT)", International Journal of Science and Research (IJSR), Volume 8 Issue 7, July 2019.

[18] Dhakad Kunal, et al.,"Smart Home Automation using IOT", International Journal of Advanced Research in Computer and Communication Engineering, Vol. 5, Issue 2, February 2016.

[19] K. Dhanusree, et al.,"An IoT Based Smart Home Security and Home Automation System", International Journal of Trend in Scientific Research and Development (IJTSRD), Volume 4 Issue 2, February 2020.

[20] Jetendra Joshi, et al.,"Performance enhancement and IoT based monitoring for smart home", pub. in proc. of IEEE International Conference on Information Networking (ICOIN), pg 468-473, 2017.

Table 1. A comparative review of the related work onto IoT based Smart Homes

Sr. No.	Author Name	Issues Addressed	Implement ation	Key points	Merit	Demerit	References
1	E. N. Ganesh (2017)	GSM Technology Implementation of IOT Architecture for SMART HOME	Real Time Implementat ion	Offers a system that would allow users to operate and monitor smart gadgets through the internet.	Provides a web-based IoT architecture based on GSM for the deployment of smart home apps, as well as a GSM-based smart house control system design.	Data transmission and authentication processes need to be made more secure.	[1]
2	Laila S., et al (2016)	IoT-Based Smart Home with Low Energy Consumption	Real Time Implementat ion	An energy-efficient IoT-based smart house is demonstrated using a home model.	Antenna modules are unique to each wireless device + Antenna module on a smart LED, HVAC system	Not Addressed	[2]
3	Nasir N.K., et al (2020)	IoT Implementation Using Android SDK and Cayenne Web Server	Real Time Implementat ion	offers a smart home implementation that can regulate the house's atmosphere by analysing customer behaviour using data acquired from numerous sensors	deployment of a smart house that focuses on energy conservation	Cameras may be utilised to keep an eye on what's going on inside the house and provide a live stream + It is also possible to automate landscape watering systems.	[3]
4	Guo-Ming S., et al (2019)	Smart home system based on IoT with a mobile phone app and a virtualized cloud server.	Real Time Implementat ion	A mobile phone app with a virtualized cloud server power an IoT-based smart home system.	to finish data analysis and data management fast and properly, the suggested system blends software and hardware	Not Addressed	[4]
5	Priya N., et al (2018)	Using the Smart Door Authentication Algorithm, secure smart home automation based on IOT	Real Time Implementat ion	putting in place a mechanism to assist the elderly and disabled	The layout is simple and straightforward.	Not Addressed	[5]

6	Shivani J., et al (2020)	Comfortable Smart Home an IoT-based home automation system	Framework	to connect household and appliance technologies such as the Internet of Things and Machine Learning in a cost-effective manner in order to make them smart.	Users will be able to get timely message notifications thanks to the mobile application's connection.	Voice assistants, emergency triggers, and music/tone recognition controls for physically challenged persons may all be added to the system.	[6]
7	Satyendra K. V., et al (2019)	Home automation system that is both clever and energy efficient	Framework	a smart, energy-efficient home automation system	This article proposes a smart, energy-efficient home automation system that can access and operate home equipment from anywhere on the planet.	The suggested system needed to be tested before it could be used in a real-world setting.	[7]
8	Anitha A., et al 2017)	Home automation system	Real Time Implementation	The internet of things is being used to create a home security system.	The proposed technology may be employed in both industrial and commercial settings.	An RFID scanner may be added to the system to further enhance it.	[8]
19	Shubham N. S., et al (2018)	Arduino-based IoT Home Automation System	Real Time Implementation	A mobile host controller and numerous client modules are included in this home automation system.	People can automate the control of their electrical appliances and set up remote control.	does not provide multiple appliance control or automated fault detection in the controlled appliance	[9]
10	Mohammad A., et al (2019)	Smart device-based IOT home networking and architectural design	Real Time Implementation	connection between machines	A web browser may control the entire system.	Not Addressed	[10]
11	Amr Jadi, et al (2020)	Using IoT devices, a strategy for predicting uncertainty in the home environment is suggested.	Real Time Implementation	Sensors (IoT devices) for forecasting uncertainty	A neural network-based system is suggested as a novel technique for predicting unpredictable events in the home environment.	Not Addressed	[11]
12	S. LAKSHMI, et al (2018)	Smart Home Security System with Door Alert Powered by IoT	Framework	Using Raspberry Pi to control illegal entry to our doors	Unauthorized entrance to homes and workplaces is prevented using a Raspberry Pi connected to a web camera and a PIR motion sensor.	The proposed system has to be validated and tested on a wide scale.	[12]
13	Nazmul H.,et al (2018)	Smart Home Security Framework based on IoT	Real Time Implementation Model	Home automation system based on Cisco's IoT security architecture	The home automation system may connect through the internet and employs portable devices as a user interface.	Not Addressed	[13]
14	Lalit M. S., et al (2018)	Home automation using Arduino and the IOT	Real Time Implementation	uses an Arduino microcontroller to create a low-cost, adaptable, and dependable home automation system with added security.	Any Wi-Fi network can be used to govern the proposed wireless network.	The proposed system has to be validated and tested on a wide scale.	[14]
15	M. Poongothai, et al (2018)	Implementation of an Internet-of-Things (IoT)-based Smart Laboratory	Framework	The main goal of the proposed work is to make electrical appliances more accessible to users through an Android app and a Node-RED dashboard	Users may operate equipment such as lights, fans, projectors, and air conditioners simply by providing orders via a smart phone or the Node-RED Dashboard.	Not Addressed	[15]
16	Chakib B., et al (2014)	Smart Grid with IoT	Framework	to look at the IoT-based SG's security concerns and challenges	examined the SG's key security concerns and problems	Not Addressed	[16]

17	Pranay P. D., et al (2018)	IoT-based smart home automation security	Real Time Implementation	to regulate other electrical appliances and monitor the temperature in the house	Wi-Fi has been successfully tested on a variety of smartphones, demonstrating its mobility and adaptability.	Not Addressed	[17]
28	Kunal D., et al (2016)	Using Home security automation based on the IoT to assist the disabled and elderly.	Real Time Implementation	Using Home security automation are designed to help the aged and handicapped	The suggested solution is both secure and energy efficient.	Not Addressed	[18]
19	K. Dhanusree, et al (2020)	Wireless security Internet-of-things-based smart home automation	Real Time Implementation	Proposed a wireless home automation system for managing the features and functions of home appliances using computer and mobile devices.	Users can use their mobile devices to control their electrical equipment by programming actions.	Not Addressed	[19]
20	Jetendra J., et al, (2017)	Smart Home Performance Enhancement and Monitoring through the Internet of Things	Framework	suggested a hierarchical smart home service architecture based on common home interface devices	High-tech smart home systems are intended to be converted into simple and easy-to-use home automation systems through the use of the suggested hierarchical design.	Not Addressed	[20]

Table 2. A tabular analysis of the Communication interfaces used in Smart Home Automation

Sr. No.	Author Name	Communication	User Interface	Controller	Storage	Tools Used.	References
1	E. N. Ganesh (2017)	GSM	Web-based, mobile app	Wi- Fi–ZigBee	Cloud server, Bottle web server, MYSQL Database	ZigBee coordinator node+GSM SMS module+On the remote control centre PC, SQL is executing.	[1]
2	Laila S., et al (2016)	Wifi	Webbased	ANSYS products	Web server	ANSYS products+energy control device with three bands (2.45 GHz, 5.8 GHz, 900 MHz)+a surveillance camera and a security and motion sensor(5.8 GHz)+Light bulbs made with LEDs	[2]
3	Nasir N.K., et al (2020)	cloud-based storage	web-based, mobile app	Arduino nano, Cayenne android application	MYSQL server	R305 Finger Print sensor+ATmega328+DHT11 sensor.	[3]
4	Guo-Ming S., et al (2019)	Wi-Fi, ZigBee	web-based, mobile app	ZigBee wireless communication protocol with Raspberry Pi +virtualized cloud server + mobile phone app.	MYSQL server	Raspberry Pi ARM Cortex-A53+1GB LPDDR2 RAM with the Linux operating system+	[4]
5	Priya N., et al (2018)	Wi-Fi	web-based, mobile app	Arduino UNO	Not Addressed	ESP8266 Wifi Module+	[5]
6	Shivani J., et al (2020)	Wifi	web-based, mobile app	Raspberry Pi	Not Addressed	Wi-Fi with the microcontroller Node MCU	[6]
7	Satyendra K. V., et al (2019)	Wifi	web-based, mobile app	Adafruit & Arduino Software(IDE)	Not Addressed	NodeMcu(ESP8266)+Adafruit & Arduino Software (IDE)	[7]
8	Anitha A., et al 2017)	WiFi module	web-based, mobile app	Blynk App, Arduino IDE using C++ programming language.	MYSQL server	Arduino Uno+D19WiFi module, ESP8266	[8]
19	Shubham N. S., et al (2018)	Bluetooth enabled	web-based, mobile app	Arduino	MYSQL and Mobius server	Arduino+Bluetooth Module HC0 +RM0703 MQ2 gas sensor	[9]
10	Mohammad A., et al (2019)	Wifi	Webbased	WSN nodes+ Raspberry pi	Not Addressed	Raspberry pi	[10]

11	Amr Jadi, et al (2020)	Wifi	Based on Neural Network application in Web Servers	WSN nodes with ZigBee & Wi-Fi,	Not Addressed	ZigBee & Wi-Fi	[11]
12	S. LAKSHMI, et al (2018)	Wifi	web-based, mobile app	Raspberry Pi	Not Addressed	Raspberry Pi+B boards+PIR sensor	[12]
13	Nazmul H.,et al (2018)	Wi-Fi	Webbased	Cisco Kinetic IoT Platform	Not Addressed	Motion Sensor, Server-PT, Switch-2960, Laptop-PT, SBC-PT	[13]
14	Lalit M. S., et al (2018)	Wi-Fi network	web-based, mobile app	Arduino UNO microcontroller	Not Addressed	esp8266-01 with Arduino UNO microcontroller	[14]
15	M. Poongothai, et al (2018)	Wi-Fi network	web-based, mobile app	Arduino UNO microcontroller	Not Addressed	ESP8266+Arduino UNO ATmega328P +Current Trans-former + Relay	[15]
16	Chakib B., et al (2014)	Wi-Fi network	Web based	Zigbee v1	Not Addressed	Not Addressed	[16]
17	Pranay P. D., et al (2018)	Wi-Fi network	web-based, mobile app	ARDUINO-UNO(microcontroller).	Not Addressed	Arduino IDE+ESP8266 chip	[17]
28	Kunal D., et al (2016)	Wi-Fi network	web-based, mobile app	Arduino microcontroller	MYSQL server	Arduino micro-controller AT mega2560+ASP.NET	[18]
19	K. Dhanusree, et al (2020)	Wi-Fi network	web-based, mobile app	Arduino microcontroller	MYSQL and Mobius server	Arduino IDE	[19]
20	Jetendra J., et al, (2017)	Wi-Fi network	web-based, mobile app	Raspberry Pi and Arduino with NRF modules integrated	Not Addressed	Raspberry Pi and Arduino with NRF modules +USB cables, DIO lines, and RS485 wiring schemes	[20]